Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2021
Course Title: Chemical Spectroscopy
Course: Chem. 340 Full Mark: 70 Time: 4 hours

(*Answer any five questions*)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
   (b) Mention the name of different regions of spectrum according to the increasing frequency of the regions. 2
   (c) What is meant by quantization of energy? Illustrate how all forms of energies are quantized. 1+2= 3
   (d) Draw and label a block diagram of single beam spectrophotometer and describe each component mentioning its function. 3
   (e) The wavelength associated with a photon is 500 nm. Calculate its wavenumber, frequency and energy [Given that, $c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum: H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6? 2
   (b) Classify the following molecules on the basis of the principle moment of inertia: HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH. 2
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
   (d) What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
   (e) Calculate the moment of inertia of water molecule around the axis defined by bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Chemical Spectroscopy
Course # Chem. 340, Full Marks: 70, Time: 4 hrs.

[Answer any five questions]

1. (a) What are meant by induced absorption and stimulated emission processes? Illustrate these processes with energy diagram and example. 1.5+1.5
(b) State the characteristics of electromagnetic radiation. Write the different regions of spectrum according to the increasing frequency of the regions. 1.5+0.5
(c) Illustrate the different factors which influence the intensity of spectral lines. 3.0
(d) Sketch a diagram of a grating spectrometer. State the purposes of the use of slits in grating spectrometer. 1+1
(e) What is meant by Doppler broadening? How could you remove Doppler broadening? 4.0

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 1+2
(b) Mention the limitations of microwave spectroscopy. 2.0
(c) In the rotational spectrum of CO at 25° C, the $J = 3 \rightarrow J = 4$ transition is more intense than the $J = 0 \rightarrow J = 1$ transition. Explain. 3.0
(d) Explain the reason why the rotational constants of $^{12}C^{16}O$ and $^{13}C^{16}O$ are 1.92118 cm^{-1} and 1.83669 cm^{-1}, respectively. 2.0
(e) The first rotational line in the rotational spectrum of CO is observed at 3.84235 cm^{-1}. Calculate the moment of inertia (I) and bond length of CO. The relative atomic weight C = 12.00 and O = 15.9994, the absolute mass of H = 1.67343×10^{-27}kg. 4.0

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part III Final Examination 2018
Course Title: Chemical Spectroscopy
Course No. Chem. 340; Full Mark: 70
Time: 4 hours

(Answer any five questions)

1. (a) What is electromagnetic radiation? Mention its important characteristics. 3
(b) Arrange different types of electromagnetic radiation with increasing order of energy. 3
(c) Why does a molecule produce a band spectrum but an atom a line spectrum? 2
(d) Discuss important factors that contribute to intensity of spectral lines. 3
(e) Calculate the energy associated with 1.00 mol of photons of green light (555 nm) [$c = 3.00\times10^{10}$ cm s^{-1}; $h = 6.626\times10^{-34}$ Js]. 3

2. (a) Which of the following molecules will show a microwave rotational spectrum? 2
H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6
(b) Classify the following molecules on the basis of the principle moment of inertia: 2
HCl, CO_2, SiH_4, SF_6, NH_3, CH_3Cl, H_2O, CH_3OH.
(c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator model. Show that the absorption spectrum of a rigid rotator is expected to consist of a series of equidistant lines. 3
(d) What are the limitations of the rigid rotator model? Write down the energy expression for non-rigid rotator in cm^{-1}. Compare the energy spacing between these two models. Discuss the isotope effect on rotational spectra. 4
(e) Calculate the moment of inertia of water molecule around the axis defined by bisector of HOH bond. Bond angle (HOH) = 104.5° and bond length (OH) = 95.7 pm. (H=1.0079, atomic mass unit = 1.6606×10^{-27} kg). 3

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) (Part III) Final Examination 2017
Course Title: Chemical Spectroscopy
Course No. Chem. 340F
Time: 4 Hours, Total Marks: 70

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[Answer any five questions]

1. (a) What are the differences between induced absorption and stimulated emission processes? 2

(b) Mention the characteristics of electromagnetic radiation. Mention the name of different regions of spectrum according to the increasing frequency of the regions. 2

(c) Sketch a diagram of a grating spectrometer. Mention the purposes of the use of slits in grating spectrometer. 2.5

(d) What is meant by quantization of energy? Illustrate how all forms of energies are quantized. 2.5

(e) State the factors which cause the broadening of spectral lines. Explain how molecular effusive beam can be produced. Describe effusive molecular beam method to remove pressure broadening. 2+1+2

2. (a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 2

(b) Mention the limitations of microwave spectroscopy. 1.5

(c) Describe briefly the effect of different factors on the rotational energy and spectrum. 3

(d) Discuss the mechanism of action of a microwave oven. What are the differences between microwave oven and conventional oven? 2+1

(e) Write the energy expression and show allowed transitions between rotational energy levels for rigid diatomic molecule. Also show that absorption spectrum of rigid diatomic molecule consists of lines at 2B, 4B, 6B, 8B, ----- cm^{-1}. 1.5+1.5

(f) The first line of carbon monoxide molecule for the transition from J=0 to J=1 was observed to be as 3.84235 cm^{-1}. Calculate the rotational constant (B) and moment of inertia (*I*). 1.5

Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Chemical Spectroscopy
Course No: Chem - 340
Time: 4 Hours, Total Marks: 70

Answer any five questions

1. (a) What is meant by induced absorption? Differentiate between spontaneous emission and stimulated emission.. 1+1.5
   (b) State the characteristics of electromagnetic radiation. 1.5
   (c) What is your concept of quantization? Explain why any form of energies of a molecule is quantized. 1+2.5
   (d) How could you remove Doppler line broadening with the help of Lamb dip spectroscopic method? 4
   (e) Sketch the spectra for the following: 2
   (i) ^{1}H-NMR spectrum of ethanol and
   (ii) Ultra-violet spectrum of propanone. 2
   (f) What are the purposes of the use of slits in grating spectrometer? 2
2. (a) Which of the following molecules will show a microwave rotational spectrum: 2
   H_2, HCl, CH_4, CH_3Cl, CH_2Cl_2, H_2O, SF_6?
   (b) Classify the following molecules on the basis of the principle moment of intertia. 2
   HCl, CH_4, CH_3F, H_2O, OCS, BCl_3
   (c) Write down the energy expression in cm^{-1} for a diatomic molecule based on rigid rotator. Show that the absorption spectrum of the rigid diatomic molecule is expected to consist of a series of equidistant lines. 3
   (d) (i) Write down the energy expression in cm^{-1} for a non-rigid rotator. Compare the change in rotational energy levels and spectrum when passing from a rigid to a non-rigid diatomic molecule with the help of a figure. 0.5+2
   (ii) explain the effect of isotope substitution on rotational spectra of a diatomic molecule such as coarbon monoxide with the help of appropriate figures and equations. 1.5
   (e) The rotational spectrum of $^{79}B.^{19}F$ shows a series of equidistant lines 0.71433 cm^{-1} apart. Calculate the rotational constant B, and hence the moment of inertia and bond length of the molecule.. 3

Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: Chemical Spectroscopy
Course No: Chem - 340
Time: 4 Hours, Total Marks: 70

Answer any five questions

a) Explain stimulated emission process with example. Differentiate between induced absorption and stimulated emission process. 1+2

b) What are meant by Doppler and pressure broadening? Describe effusive molecular beam method to remove line broadening. 2+1+2

c) Sketch a diagram of grating spectrometer. Explain why sample is placed near the exit slit rather than entrance slit in the grating spectrometer. 2.5

d) State the factors which cause the broadening of spectral lines. 2.0

e) What do you mean by signal to noise ratio? 1.5

a) State the criteria for a molecule to be microwave active. Give examples of at least four molecules which may show rotational microwave absorption spectrum. 2.0

b) Mention the limitations of microwave spectroscopy. 2.0

c) What is rotational constant (B)? Describe briefly the effect of different factors on the values of B. 4.0

d) Explain the schematic diagram why rotational energy levels become lowered when passing from rigid to non-rigid diatomic molecules. 4.0

e) Write the energy expression and show allowed transitions between rotational energy levels for a rigid diatomic molecule. 2.0